INCH-POUNDS

MIL-C-16839E(OS)
25 October 1990
SUPERSEDING
MIL-C-16839D(OS)
29 May 1969

MILITARY SPECIFICATION

CABLE, SPECIAL PURPOSE, ELECTRICAL (UNDERWATER USE)

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers single and multi-conductor special purpose electrical cable for underwater use.

1.2 Classification. Cable shall be of the following types, as specified (see 6.2):

Type DMU	Double Conductor, demolition, magnetic
Type DNU	Double Conductor, demolition, nonmagnetic
Type DPU	Double Conductor, oil resistant, portable
Type MSU	Multiple Conductor, steel core
Type MTU	Multiple Conductor, cotton or jute core
Type SPU	Single Conductor, heat and oil resistant flexible

1.3 Conductor size. The conductor size shall be in accordance with American Wire Gauge (AWG) size to 0000AWG; larger sizes shall be indicated by circular mil area.

1.4 Military part number. Military part numbers shall be in accordance with the following example:

M 16839 - DMU 4

Where:

M is a prefix meaning military specification
16839 is the specification number
DMU is the cable type
4 is the AWG size

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6145

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-W-343 Wire, Electrical and Nonelectrical, Copper (Uninsulated)

MILITARY

MIL-C-915 Cable and Cord, Electrical, for Shipboard Use, General Specification for

MIL-W-83420 Wire Rope, Flexible, For Aircraft Control

STANDARDS

FEDERAL

FED-STD-228 Cable and Wire, Insulated; Methods of Testing

(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM A 228 Standard Specification for Steel Wire, Music Spring Quality (DOD Adopted)

ASTM A 818 Standard Specification for Coppered Carbon Steel Wire (DOD Adopted)

ASTM D 2000 Standard Classification System for Rubber Products in Automotive Applications (DOD Adopted)

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(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents may also be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Unless otherwise specified, all cables shall be manufactured in accordance with the requirements of MIL-C-915.

3.2 Materials.

3.2.1 Conductors.

3.2.1.1 Copper. Copper conductors shall be in accordance with QQ-W-343, except that the inclusion of bismuth in the tin-lead alloy coating is optional.

3.2.1.2 Other conductors. Conductors of other materials shall be as specified in 3.4.

3.2.2 Strength members. Members to increase the cable strength shall be as specified in 3.4.

3.2.3 Insulation, natural rubber.

3.2.3.1 Application. Each conductor shall be covered for its entire length with a well-centered wall of vulcanized rubber.

3.2.3.2 General composition. The insulation compound shall be a homogeneous mixture. The rubber shall be in accordance with line call-out 4AA520A13C20EA14J of ASTM D 2000. The compound shall contain only the following ingredients: rubber, sulphur, inorganic mineral filler, waxy hydrocarbons, organic accelerators, and anti-oxidant. The sulphur shall be of the highest purity and of extreme fineness. The inorganic mineral fillers shall be of extreme fineness and shall be substantially free of moisture, red lead, free carbon, or any substance likely to exert a deleterious effect on the rubber. The waxy hydrocarbons shall be free from saponifiable matter.

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3.2.3.3 Chemical analysis. Unless otherwise approved, insulation removed from finished conductors shall conform to the following chemical requirements (percentages are shown by weight):

Rubber	Not less than 40 percent
Waxy hydrocarbons	Maximum 3 percent
Total sulphur	Maximum 3 percent
Free sulphur	Maximum 0.5 percent
Inorganic mineral fillers, organic accelerators, and antioxidants	Remainder
Specific gravity	Between 0.93 and 1.14

3.2.3.4 Physical properties. Insulation removed from conductors not earlier than 48 hours after completion of the cable shall meet the following physical requirements at ordinary room temperature, 65 to 90 degrees Fahrenheit (°F) (18 to 32 degrees Celsius (°C)):

a. The tensile strength shall be not less than 1300 pounds per square inch of original cross-sectional area.

b. The set shall be not more than 3/8 inch in 2 inches after testing in accordance with 4.3.1.1.1.

c. After testing in accordance with 4.3.1.1.1, the specimen shall undergo a second elongation from the "length after set" to at least 10 inches, prior to rupture.

d. After 96 hours in an atmosphere of oxygen at 300 pounds per square inch pressure at a temperature of 158°F (70°C), the maximum depreciation of tensile strength and elongation shall not exceed 25 percent. A maximum variation of 10 pounds per square inch, 2°F (1°C), and 1 hour, in either direction during the aging process, is permissible.

3.2.4 Sheath.

3.2.4.1 Application. The sheath compound shall be applied concentrically about the grouped conductors and so cured as to assure their accurate centering within the finished sheath. The finished, sheathed cable shall be circular in cross section and of uniform diameter throughout its length.

3.2.4.1.1 Curing. The compound shall be cured in a close fitting mold or by the continuous extrusion method. After vulcanizing, the sheath shall be a firmly bound, strong, highly elastic, homogeneous mass.

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3.2.4.2 Physical properties. Sheathing removed from finished cables not earlier than 48 hours after vulcanization, shall meet the following physical requirements at ordinary room temperature, 65 to 90°F (18 to 32°C):

a. The tensile strength shall be not less than 3500 pounds per square inch of original cross-sectional area.

b. The set shall be not more than 3/16 inch in 2 inches after testing in accordance with 4.3.1.1.1.

c. After testing in accordance with 4.3.1.1.1, the specimen shall undergo a record elongation from the "length after set" to at least l0 inches, prior to rupture.

d. After 96 hours in an atmosphere of oxygen at 300 pounds per square inch pressure at a temperature of 158°F (70°C), the maximum depreciation of tensile strength and elongation shall not exceed 25 percent. A maximum variation of 10 pounds per square inch, 2°F (1°C), and 1 hour, in either direction during the aging process, is permissible.

3.2.4.3 Surface finish. The surface of the finished cable shall present a smooth, dense, shiny appearance free of all roughness or other regularities. There shall be no objectionable waviness due to the failure of the sheath to tightly grip the conductors.

3.3 Manufacturer's identification. A marker tape in accordance with MIL-C-915 shall be provided for each cable except that the required information shall be as follows:

Specification MIL-C-16839
Military part number. For type DMU and DNU, include whether magnetic or nonmagnetic
Name of manufacturer
Name and location of plant
Year of manufacture

3.4 Detail requirements. Detail requirements or exceptions applicable to individual types are specified in table I and in 3.4.1 through 3.4.6.4.

3.4.1 Type DMU.

3.4.1.1 Construction.

First. The sheath compound shall be a homogeneous mixture. The rubber shall be in accordance with line call-out 4AA535A13C20EA14J of ASTM D 2000.

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TABLE I. Physical and electrical requirements.

Requirement	Unit of Measurement	Type of Cable					
		DMU-4	DNU-4	DPU-10	MSU-21	MTU-7	SPU-500
Operating voltage	volts, ac, max volts, dc, max	300 500	300 500	600 1000	600 1000	600 1000	600 1000
Conductor size	strands	7 steel and 12 copper	19	19	7	41	5054 (19 x 7 x 38)
	diameter, inches	0.0142	0.0142	0.0234	0.0242	0.0063	0.010
Insulation thickness	inches, min	0.0156	0.0156	0.047	0.05	0.031	0.117
Sheath thickness	inches, min	0.04	0.04	0.06		0.125	0.12
Inner sheath thickness	inches, approximate				0.13		
Total sheath thickness	inches, min				0.26		
Outside diameter	inches	0.34 ± .01	0.34 ± .01	0.596 ± .015	1.73 ± .03	0.56 ± .015	1.505 ± .015
Conductor resistance	ohms/1000 feet at 68°F (20°C) max	4.3	4.3	1.05	2.73	7.15	0.023 at 25°C
Insulation resistance	megohms/1000 feet, min	100	100	500	500	300	90
Dielectric strength	volt, min	1500	1500	2500	2500	2500	5000
Abrasion resistance	number of revolutions of wheel without wearing through sheath	50	50				
Air leakage	cubic inches			0	0	0	0
Water leakage	cubic inches			0	0	0	0

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Second. A conductor of coated wire in accordance with 3.2.1.1. Successive concentric lays to be of opposite hand on the steel strength member.

Third. A separator (optional).

Fourth. A wall of natural rubber insulation (see 3.2.3). The rubber shall be color-coded for the identification of conductors. The choice of colors is optional.

Fifth. A rayon inner braid (see MIL-C-915).

Sixth. Two conductors, twisted together with a lay not greater than 3 inches, and a treated moisture-resistant jute filler free of excess wax (less than 2 percent extractable, the wax to be other than paraffin), to form an essentially round assembly.

Seventh. A binder braid over the assembly.

Eighth. A natural rubber sheath:

General composition--The sheath compound shall consist of a homogeneous mixture with the rubber content being either fine Para or best quality plantation-grown Hevea Brasiliensis rubber. The rubber shall not have been previously used in a compound. The remainder of the compound shall consist of sulphuric, inorganic mineral fillers, carbon black, anti-oxidants, and organic accelerators.

Chemical analysis--Sheathing removed from finished cables shall conform to the following chemical requirements (percentages are shown by weight):

Rubber	Minimum 60 percent
Waxy hydrocarbons	Maximum 3.3 percent
Total sulphur	Maximum 3 percent
Free sulphur	Maximum 1.2 percent
Carbon black	Minimum 20 percent
Inorganic mineral fillers, organic accelerators, and antioxidants	Remainder
Specific gravity	Minimum 1.12

3.4.1.2 Marking. The sheath shall be marked "Type DMU*--MAGNETIC DEMOLITION CABLE." This marking, to be repeated at least once per foot, shall be reasonably permanent under all normal rough handling and storage conditions and shall be printed in contrasting color on the sheath along the entire length of the cable. The above sheath marking is in addition to the marking tape called for in 3.3.1. (* For application size number see 1.3.)

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3.4.2 Type DNU.

3.4.2.1 Construction.

First. A conductor of tin or tin-lead alloy coated cadmium bronze wire of 85 percent conductivity grade; 80,000 pounds per square inch minimum tensile strength. Successive concentric lays to be of opposite hand.

Second. A separator (see MIL-C-915).

Third. A wall of natural rubber insulation (see 3.2.3). The rubber shall be color-coded for the identification of conductors. The choice of colors is optional.

Fourth. A rayon inner braid (see MIL-C-915).

Fifth. Two conductors, twisted together with a length of lay not greater than 3 inches, and a treated moisture-resistant jute filler free of excess wax (less than 2 percent extractable, the wax to be other than paraffin), to form an essentially round assembly.

Sixth. A binder braid over the assembly.

Seventh. A natural rubber sheath.

General composition--The rubber shall be in accordance with line call-out 4AA535A13C20EA14J of ASTM D 2000. The remainder of the compound shall consist of sulphuric, inorganic mineral fillers, carbon black, anti-oxidants, and organic accelerators.

Chemical analysis--Sheathing removed from finished cables shall conform to the following chemical requirements (percentages are shown by weight):

Rubber	Minimum 60 percent
Waxy hydrocarbons	Maximum 3.3 percent
Total sulphur	Maximum 3 percent
Free sulphur	Maximum 1.2 percent
Carbon black	Minimum 20 percent
Inorganic mineral fillers, organic accelerators, and antioxidants	Remainder
Specific gravity	Minimum 1.12

3.4.2.2 Marking. The sheath shall be marked "Type DNU*--NONMAGNETIC DEMOLITION CABLE." This marking, to be repeated at least once per foot, shall be reasonably permanent under all normal rough handling and storage conditions and shall be printed in contrasting color on the sheath along the entire length of the cable. The above sheath marking is in addition to the marking tape called for in 3.3.1. (* For application size number see 1.3.)

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3.4.3 Type DPU.

3.4.3.1 Construction.

First. A conductor of coated copper wire in accordance with 3.2.1.1. Successive concentric lays to be of opposite hand.

Second. A separator (see MIL-C-915).

Third. A wall of Butyl rubber (see MIL-C-915). The rubber shall be color-coded for the identification of conductors. The choice of colors is optional.

Fourth. Two conductors, twisted together with a length of lay not greater than 5 inches, and synthetic rubber fillers to form an essentially round assembly.

Fifth. A binder braid over the assembly.

Sixth. A polychloroprene sheath (see MIL-C-915).

3.4.4 Type MSU.

3.4.4.1 Construction.

Strength member.

First. A 1/4-inch-diameter (7 x 19) steel cable in accordance with MIL-W-83420, Type I, Composition B, except that there shall be no lubricating compound applied to the cable. Also acceptable, a 1/4-inch-diameter (7 x 7) steel cable, minimum breaking strength 6100 pounds, which is in accordance with MIL-W-83420, Type I, Composition B and the above exception in all other respects.

Second. Two layers of cotton braid (see MIL-C-915).

Conductor.

Third. A conductor of coated copper wire in accordance with 3.2.1.1 concentric lay.

Fourth. A wall of Butyl rubber (see MIL-C-915). The rubber shall be color-coded for the identification of conductors. The choice of colors is optional.

Fifth. An inner braid. Required only if necessary for circuit identification (see MIL-C-915).

Sixth. Twenty-one conductors, twisted around the steel strength member in successive concentric lays of suitable length and opposite hand, and fillers to form an essentially round assembly.

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Seventh. A binder braid over the assembly.

Eighth. A polychloroprene inner sheath (see MIL-C-915).

Ninth. A cotton reinforcement (see MIL-C-915).

Tenth. A polychloroprene outer sheath (see MIL-C-915).

3.4.5 Type MTU.

3.4.5.1 Construction.

First. A conductor of coated copper wire in accordance with 3.2.1.1 Bunch stranded.

Second. A separator (see MIL-C-915).

Third. A wall of Butyl rubber (see MIL-C-915). The rubber shall be color-coded for the identification of conductors. The choice of colors is optional.

Fourth. An inner braid. Required only if necessary for circuit identification (see MIL-C-915).

Fifth. Seven conductors, twisted in a concentric lay around a central core of impregnated cotton or jute, and fillers to form an essentially round assembly (see MIL-C-915). The length of lay shall be not greater than 3 inches. A conductor shall not be used in or for the central core.

Sixth. A binder braid over the assembly.

Seventh. A polychloroprene sheath (see MIL-C-915).

3.4.6 Type SPU.

3.4.6.1 Construction.

First. A conductor of uncoated copper wire, SD Class K extra flexible. Rope-lay-stranding with concentric-lay bunch stranded members.

Second. A separator of T-62 tape (opaque mylar) 4 inches by 2 mils.

Third. A primary insulation of Black Hypalon.

Fourth. A reinforcement of 2200 Den Rayon, double serve in reverse, 2-inch lay.

Fifth. A Black Hypalon sheath.

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3.4.6.2 Controlling specifications. The cable shall be extra flexible multiple strand with special insulation (UL Type RHH-RHW 600 Volt alternating current/1000 Volt direct current) and have a current-carrying capacity of 730 amperes at 40°C. The cable shall have a temperature rating of 105°C.

3.4.6.3 Cold bend resistance of sheath. The sheath shall not crack at -22°F (-30°C) when the cable is bent two turns around a 3-inch mandrel.

3.4.6.4 Sheath marking. Sheath marking shall be in accordance with MIL-C-915, except use CV white ink, and include the following: manufacturer's name, 500 MCM 600 Volt (UL) Type RHH-RHW, CSPE, year of manufacture.

3.5 Workmanship. The material as specified herein shall be processed as required in accordance with the best practice for a high quality cable and shall be suitable for its intended purpose.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Quality conformance inspection. Unless otherwise specified in the contract or order, only lot quality conformance tests shall be required. Tests required for each cable type shall be as specified in tables II and III, applicable documents, and as stated below. Unless otherwise specified in the following paragraphs, all sampling and testing shall be in accordance with MIL-C-915, as applicable.

4.2.1 Sample size. The sample to be subjected to the physical tests shall consist of three specimens of sufficient length selected at random from each lot of completed cable.

4.2.2 Action in case of failure.

4.2.2.1 Inspection of product and electrical tests. Each continuous length of cable failing any of these tests shall be rejected.

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TABLE II. Quality conformance tests required on sample lengths.

Test Title	Source of Test	Type of Cable					
		DMU-4	DNU-4	DPU-10	MSU-21	MTU-7	SPU-500
Copper strand area, tensile strength and coating	QQ-W-343	x					
Steel strand area, tensile strength and coating	ASTM A 818 ASTM A228/A 228M83	x					
Strand area, tensile strength and coating	QQ-W-343		x [1]		x		
Strand area and tensile strength	QQ-W-343			x		x	x
Insulation and sheath properties [2]	4.3.1	x	x				
Insulation and sheath properties [2]	MIL-C-915			x	x	x	x
Hot oil test	MIL-C-915			x			x
Water and air leakage	4.3.3			x	x	x	x
Flammability	MIL-C-915						x
Cold bond	MIL-C-915						x

[1] See "Construction, First" of 3.4.2, Type DNU.

[2] Chemical analysis to be performed on initial lot only or when specifically requested by the Government inspector.

TABLE III. Quality conformance tests required on each continuous length.

Test Title	Source of Test	Type of Cable					
		DMU-4	DNU-4	DPU-10	MSU-21	MTU-7	SPU-500
Inspection of product	MIL-C-915	x	x	x	x	x	x
Conductor resistance	FED-STD-228, Method 6021	x	x	x	x	x	x
Insulation resistance	MIL-C-915	x	x	x	x	x	x
Dielectric strength	MIL-C-915	x	x	x	x	x	x
Abrasion resistance	4.3.2	x	x				

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4.2.2.2 Physical tests. Failure of any specimen in any of these tests shall cause rejection of the represented lot. Lots found unacceptable shall be resubmitted for Government inspection only after all units are reexamined and retested and all defective units are removed or defects corrected. Resubmitted lots shall be kept separate from new lots.

4.3 Inspection methods.

4.3.1 Insulation and sheath test. Testing for physical and chemical properties of insulation and sheath materials shall be in accordance with MIL-C-915, except as follows.

4.3.1.1 Natural rubber. See 3.2.3; 3.4.1.1 Construction, Eighth; and 3.4.2.1 Construction, Seventh.

4.3.1.1.1 Set test. A test specimen, having a length of not less than 6 inches, marked with bench marks 2 inches apart shall be placed in the jaws of a testing machine and stretched until the bench marks are 6 inches apart for both the 40 and 60 percent rubber compounds (see 3.2.3; 3.4.1.1 Construction, Eighth; and 3.4.2.1 Construction, Seventh). After the termination of stretch, the test specimen shall be released within 5 seconds and the set determined 1 minute after the beginning of release.

4.3.2 Abrasion resistance test. The specimen shall be a 2 foot length of completed cable.

4.3.2.1 Procedure. The specimen shall be hung with a 90 degree arc of contact over a 7-inch-diameter abrasion wheel make up of twelve 3/8-inch-diameter fine grade tungsten carbide rods and two 7-inch-diameter phenolic discs arranged in the form of a squirrel cage. A 15 pound weight shall be attached to the free end of the specimen. The tests shall be made at room temperature with the abrasion wheel rotating at 60 revolutions per minute (rpm) in the direction which tends to lift the 15 pound weight. The number of wheel revolutions required to wear through a measured thickness of the sheath wall shall be determined and recorded.

4.3.3 Water and air leakage. The specimen shall be a 30 inch section of completed cable with or without end seal.

4.3.3.1 Apparatus. The apparatus shall consist of a testing tank, 15 inches long and suitable diameter to accommodate the test specimen, fitted with Navy Standard watertight stuffing tubes with approved packing, connected to the pressure system, a pressure control and leakage detection device permitting test by either air or water under pressure, and necessary connections, gages, and valves between testing tank and pressure control system.

4.3.3.2 Procedure. The specimen shall be sealed in place into or through the testing tank. Air pressure shall be applied to detect leakage through the cable sheath or end seal, and to enable location and correction of leaks through stuffing tubes and packing. The cable sheath or end seal and the gland shall not leak under this air pressure. After the assembly has withstood the air pressure for 30 minutes without leakage, the specimen shall be subjected to water pressure at room temperature for 4 hours. The amount of water leakage, if any, shall be determined to a precision of 1 cubic inch.

4.3.3.3 Pressures. The air and water pressures applied shall be 75 pounds per square inch (lb/in^2).

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5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5.1 Packaging, packing, and marking. The cable length specified in 5.2 shall be packaged, packed, and marked in accordance with the requirements of MIL-C-915, as applicable.

5.2 Cable length. All cable shall be furnished on reels in standard lengths as follows:

a.	Type DMU and DNU	1000 ± 50 feet
b.	Type DPU	2000 ± 200 feet, except that a maximum of 30 percent of the total order may be in shorter lengths, as follows:
	900 to 1800 feet	not more than 30 percent of the total order
	450 to 900 feet	not more than 10 percent of the total order
	200 to 450 feet	not more than 5 percent of the total order
c.	Type MSU	1000 ± 50 feet, except that a maximum of 10 percent of the total order may be in 500 to 950 foot lengths.
d.	Type MTU	1000 ± 50 foot lengths
e.	Type SPU	600 continuous feet +10, -0 percent with no random or remnant lengths.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. All cable covered by this specification is suitable for underwater use. Primary uses for the various types are as follows:

Type DMU	For firing explosives. High tensile strength. Magnetic characteristics. Not for permanent installation.
Type DNU	For firing explosives. Medium tensile strength. Nonmagnetic characteristics. Not for permanent installation.
Type DPU and MSU	For degaussing range installation.
Type MTU	For degaussing magnetometer installations.
Type SPU	For deperming use.

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Type and size of cable required (see 1.2 and 1.3)

c. Military part number required (see 1.4)

d. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2)

e. Quantity

6.3 Subject term (key word) listing.

Conductors
Demolition
Nonmagnetic
Rubber insulation
Sheath
Wire rope

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS

Review Activity:
DLA - IS

(Project 6145-N340)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-C-16839E(OS)	25 OCTOBER 1990

3. DOCUMENT TITLE
CABLE, SPECIAL PURPOSE, ELECTRICAL (UNDERWATER USE)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
ELLIOTT, ROLLAND V.	301-743-4510/4358	364-4510/4358

c. ADDRESS (Include Zip Code)
COMMANDING OFFICER
NAVAL ORDNANCE STATION (3730D)
INDIAN HEAD, MD 20640-5000

IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:
Defense Quality and Standardization Office
5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466
Telephone (703) 756-2340 AUTOVON 289-2340

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